



Habitat preference and functional traits influence responses of tropical butterflies to varied habitat disturbance

Suman Attiwilli^{1,2} · Tarun Karmakar² · Kavita Isvaran¹ · Krushnamegh Kunte²

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Abstract

Biodiversity is declining worldwide due to anthropogenic disturbances. There is a vital need to be able to predict the impacts of these disturbances on ecological communities. Species' attributes such as habitat preference and functional morphological traits may help in uncovering broad patterns in their responses to anthropogenic habitat disturbance. We examined trait-habitat associations in 251 tropical butterfly species by measuring three morphological traits (body mass, wing length and proboscis length) and classifying species according to their habitat preference (forest, ecotone, open). We then correlated these measures (morphological traits and habitat preference) to responses of butterflies to two common habitat disturbances, i.e., a road passing through forest, and dominance of an invasive plant *Lantana camara*. Both evolutionary relatedness (i.e., family affiliation) and habitat preference explained considerable variation in butterfly morphological traits, suggesting a relationship between evolutionary relatedness, habitat preference and morphology. Both habitat preference and morphological traits influenced response to habitat disturbance: (a) forest roads were frequented by smaller butterflies (b) *Lantana*-dominated patches were depauperate in larger butterfly species and species with long probosces that are usually associated with deep-flowered forest plants, and (c) habitat preference was related to disturbance from both road and *Lantana*, with forest-specialized butterfly species showing a negative response. These results suggest that tropical butterflies show generalized responses to habitat disturbance according to their habitat preference and functional morphological traits.

Keywords Invasive species · Roads · *Lantana camara* · Micro-habitats · Morphological variation · Allometry

Introduction

A major challenge in ecology is to understand changes in species abundance through space and time. This has become especially urgent, considering the impacts of anthropogenic disturbances on the biosphere (Vitousek et al. 1997; Dirzo and Raven 2003; Wagner 2020). Recently, trait-based approaches have been advocated to derive general patterns and predict changes in species composition along environmental gradients (McGill et al. 2006; Webb et al. 2010; Cadotte et al. 2015; Wong et al. 2019). Traits can also be useful to quantify responses of ecological communities to

anthropogenic disturbances (Mouillot et al. 2013). Species with particular traits might reduce in abundance or filter out due to disturbance; thus, trait-based indices could present a more consistent picture of impact than taxonomic indices (Mouillot et al. 2013).

A broad set of morphological, ecological and life-history traits have been used to relate responses of ecological communities to anthropogenic disturbances such as climate change, land use change and fragmentation (Hambäck et al. 2007; Barbaro and Van Halder 2009; Öckinger et al. 2010; Vandewalle et al. 2010; Diamond et al. 2011; Newbold et al. 2013; Jain et al. 2016). In a meta-analysis of bees, traits such as nest location (above vs. below ground) and sociality (solitary vs. social) affected species responses to multiple habitat disturbances (Williams et al. 2010). A review of the response of Neotropical vertebrates to forest fragmentation identified habitat preference as a key predictor; fragmentation seemed to affect forest-dependent species more negatively than their open-habitat counterparts (Vetter et al. 2011). Larger

✉ Suman Attiwilli
sumanattiwilli@gmail.com

¹ Centre for Ecological Sciences, Indian Institute of Science, Bangalore 560012, India

² National Centre for Biological Sciences, Tata Institute of Fundamental Research, Bangalore 560065, India

sized butterflies were more vulnerable to extinction from fragmentation in a study of tropical frugivorous butterflies (Shahabuddin and Ponte 2005).

Invasive plants and linear intrusions such as roads are two widespread anthropogenic disturbances in tropical forests. Invasive plants can affect the behaviour, abundance and richness of the native faunal community by providing a novel resource, altering habitat structure or displacing native plant resources (Vilà et al. 2011; Bezemer et al. 2014; Jain et al. 2016). Roads can impact biodiversity by causing road-related mortality, fragmentation of habitat, and edge-effects (Laurance et al. 2009). In a study of dung beetle assemblages in dry savannah and an adjacent site heavily invaded with the invasive plant *Prosopis glandulosa*, larger species were excluded from the invaded site, possibly due to impairment of their flight ability and resource searching among the invasive thickets (Steenkamp and Chown 1996). In another study, which examined the impact of roads on mammal abundances, larger species appeared to show stronger declines with road densities than smaller species (Rytwinski and Fahrig 2011). A study of temperate butterflies at a landscape scale found that wing-loading ratios, indicating mobility, predicted the response of butterflies to invasive plants (Gallien et al. 2017). The linkage between species traits and responses of to these two disturbances is not adequately understood in tropical butterflies.

Butterflies are often considered as model organisms for ecological studies of insects, and as flagship species in invertebrate conservation. They exhibit a range of sizes, shapes and manner of flight, making them ideal for trait-based studies of habitat change. Body size, proboscis length and habitat associations (here defined as a preference for ‘forest’, ‘ecotone’ or ‘open’ habitats) are key niche-related traits of butterflies. Body size is a composite trait of organisms that is linked to various life-history and ecological traits such as metabolic rate, home range and abundance (Peters and Wassenberg 1983; García-Barros 2000; Brown et al. 2004; White et al. 2007). Characteristics of the proboscis, such as its structure or length, might determine which food resources butterflies can effectively utilize (Corbet 2000; Krenn et al. 2001; Kunte 2007). Habitat preferences of species reflect the micro-habitats where they live, seek mates and/or other resources, and complete their life-cycles (Willmott et al. 2017). Thus, any disturbance that changes the micro-habitat niche availability is likely to affect species differently, according to their requirements.

In this study, we examined whether functional morphological traits and habitat preferences of butterflies can help predict response to forest roads and an invasive plant *Lantana camara* (henceforth lantana) — two common habitat disturbances in tropical forests. We first describe the patterns of variation in functional morphological traits (body size and proboscis length), since there are relatively few studies on

comparative functional morphology in butterflies (Marden and Chai 1991; Dudley and Srygley 1994; García-Barros 2000). Baseline information on trait variation can be used to test various eco-evolutionary hypotheses on factors shaping the variation (Polo and Carrascal 1999; Melville and Swain 2000; Goodman 2009). For example, in a study of Nymphalid butterflies in Neotropical forests, forewing shape corresponded with aerodynamic requirements of a gliding flight in the forest understorey (Cespedes et al. 2015). We tested how habitat preference is associated with morphological traits in 251 butterfly species. We used higher-level taxonomic grouping, i.e., family affiliation of butterflies, as a crude proxy for phylogenetic non-independence, since a species-level or even a genus-level phylogeny of butterflies used in this study is currently unavailable. For the second objective, we hypothesised that a) Open habitat species would show a strong positive association with road, while forest-specialized species would show a negative association, with ecotone species showing an intermediate response b) Nectar-feeding species with either very short or very long proboscis will avoid lantana-dominated patches. Lantana has a long, slender corolla ca 1 cm deep. Species with a short proboscis may not be able to reach the nectar in lantana flowers, while species with very long proboscises, which often feed on long-flowered forest shrubs, may be competitively disadvantaged on lantana flowers (Kunte 2007). Our work showed that functional traits and habitat preferences of butterflies determine their responses to these two disturbance types, which has implications for habitat management and biodiversity conservation in tropical areas.

Methods

Morphometric measurements: We measured a single adult male of 251 butterfly species comprising five families. Greater sampling within species was not possible because of limitations of collection permits and other logistical constraints. However, within species variation is likely to be less than variation across species, hence overall trends in the measured traits may remain representative for this large pool of species. We collected the specimens from two biodiversity hotspots in India, i.e., the Western Ghats (Karnataka and Kerala) and the Eastern Himalaya (Nagaland) during multiple field visits between the months of April – October (summer and monsoon) in 2012 and 2013. We caught specimens from morning till afternoon with insect nets, and took morphometric measurements at the base camp in the same evening. We measured three traits for this study, viz., body mass and wing length (as indices for body size) and proboscis length. Body mass, i.e., weight of the whole specimen, was measured with a digital weighing-balance (SartoriusAC120S, 0.1 mg resolution). Wing length, i.e., the

longest straight-line distance from forewing base to forewing apex, and proboscis length, i.e., straight distance between the base and the tip of the proboscis, were measured using a digital calliper (Mitutoyo Digimatic Caliper CD-6" CSX, 0.01 mm resolution).

Habitat preference: We classified all species into three categories, i.e., forest, ecotone and open, according to their primary habitat preference (Kunte 2008). These are broad categories, reflecting preferred primary vegetation type and canopy openness. Forest species, according to the classification used in this paper, generally prefer evergreen, semi-evergreen, dense riparian or moist deciduous forests. Ecotone species might be generally found at the edges of evergreen forests, and in dry deciduous to secondary moist deciduous forests. Open species prefer open scrublands or dry grasslands. We excluded species that could not be classified into a single category from analysis.

Response to habitat disturbance: We observed responses of butterflies to two habitat disturbances, viz., a road passing through forest and lantana, through surveys carried out in the Biligiri Rangaswamy Temple (BRT) Tiger Reserve in the Western Ghats (Karnataka, India). Study site is a protected forest criss-crossed with unpaved roads, and heavily invaded with alien plants, primarily lantana. Studies that measured lantana occurrence and density in 1997 (Murali and Setty 2001) and 2007–2008 (Sundaram and Hiremath 2012) report that, at the landscape scale, lantana abundance increased enormously, from being present in 41% of the plots in 1997 to 81% of plots by 2008, while the density of lantana stems per plot increased threefold in the same time (Sundaram and Hiremath 2012). We demarcated a study area of 1 km × 0.3 km in the moist deciduous forest patch of BRT. To measure responses to road and lantana, we surveyed butterfly species along 100 m belt transects in two micro-habitats within the study area, i.e., the road verge ($n=10$ in “Wet” season, i.e., Sep–Oct 2012, $n=7$ in “Dry” season, i.e., Feb–Apr 2013) and the forest interior ($n=7$ in Wet season and $n=8$ in Dry season 2013). The forest interior transects were laid in areas with different levels of lantana cover. Along these interior transects, we quantified lantana extent as percentage cover, by visually estimating the percentage cover of lantana in two 5 m × 5 m plots, laid on each side of the transect every 10 m. We took the mean of these 20 readings to characterize the lantana cover for that transect. We classified transects with lantana cover < 25% as “low” lantana ($n=3$ in Wet season, and $n=4$ in Dry season), and > 25% as “high” lantana ($n=4$ in Wet season, and $n=4$ in Dry season). We walked each road and interior transect (100 m each) 3–5 times in each season, at a fixed slow pace of 10 min per walk. We obtained species abundance data by noting the identity of every butterfly encountered along the transects. To minimize identification errors and maximise detectability, we noted small understorey butterflies,

i.e., species with a wing length < 3 cm, and active near the ground to mid-canopy, in a 3 m wide belt along the transects. Bigger butterflies, i.e., those with wing length > 3 cm, and active on the ground till canopy level, were surveyed in a 10 m wide belt along the transects. Thus, small canopy-level butterflies, which are difficult to observe, were excluded from surveys. Different width transects allowed us to ensure similar detectability for large and small-sized species.

Transects thus capture small-scale behavioural responses of butterflies to the two habitat disturbances, i.e., road through forest and lantana. We used the trait database, mentioned above, to evaluate the associations between traits and response to habitat disturbance.

Statistical analysis

Relationship between functional traits, taxonomic ranking and habitat preference:

We used linear modelling to examine the association between morphological traits (body mass, wing length, proboscis length) and habitat preference, with family identity as a co-explanatory variable. Wing length and proboscis length are expected to increase allometrically with body mass, and initial explorations suggested that the relationship may vary with family identity. Hence, we ran separate linear models for wing length and proboscis length, with body mass, family, and an interaction term between family and body mass as explanatory variables, together with habitat preference. Trait values were $\log_e$ -transformed prior to running the models to satisfy assumptions about normality and homoscedasticity of residuals.

Relationship between functional traits, habitat preference and response to habitat disturbance:

Abundance of each species was pooled across a type of transect (all road vs. all interior transects; all low vs. all high-lantana transects) after dividing by the number of trials per season per transect in order to standardize for sampling effort. Data were then pooled for both seasons. To quantify response to road and high lantana, we calculated an index of a species aversion to disturbance as: $\log(A_1 + 1) - \log(A_2 + 1)$, where A_1 was its pooled abundance in the “undisturbed” habitat (i.e., forest interior/low-lantana) and A_2 was its pooled abundance in the disturbed habitat (i.e., road/high-lantana). Abundances were logged since the range of abundance values in the dataset was orders of magnitude different. “1” was added to abundances before logging to circumvent the problem of logging 0-abundance in some cases. Thus, a positive value of the index indicates an aversion to high-lantana or road, while a negative value indicates a preference for high-lantana or road micro-habitats. Note that, the use of transects of different widths (3 m or 10 m) does not affect the response measure, since abundances of each species were compared across transects of similar width.

Table 1 Number of species used for morphological measurements and their habitat associations in different families of butterflies

	Ecotone	Forest	Open	Total
Hesperiidae	11	22	9	42
Lycaenidae	13	23	5	41
Nymphalidae	28	53	8	89
Papilionidae	5	19	1	25
Pieridae	6	7	14	27

To test associations between traits and response to habitat disturbance, we ran simple linear models, with an index of aversion as the response variable, and each of the traits, i.e., body mass, wing length, proboscis length and habitat preference as individual predictors. In case of road vs. interior, ecotone and ecotone-open species were pooled into a single category, i.e., ecotone-open, due to a smaller number of species in each individual category, thus getting a factor with three levels (ecotone-open: $n = 17$; forest: $n = 17$; open: $n = 11$). In case of aversion to lantana, we pooled ecotone and open species into a single category, i.e., ecotone-open, since there were relatively few representatives of each

individual category, thus getting a factor with two levels (ecotone-open: $n = 14$; forest: $n = 16$). We did not run multiple regressions because (a) we wanted to thoroughly explore individual trait-disturbance relationships, and (b) traits were somewhat correlated and hence could not be included in a single model.

We used R statistical software (R Core Team 2016) for all analysis. An alpha level of 0.05 was used for all statistical tests.

Results

Relationship between functional traits, family and habitat preference: Sampling effort of number of species per habitat category per family are shown in Table 1. Mean and variation in morphological traits are shown in online supplementary information (SI Table 1). In the pooled dataset of families, wing length and proboscis length ranged an order of magnitude, while body mass ranged three orders of magnitude. This large variation in morphological traits appeared to vary both with family and habitat (Table 2). Family and habitat preference explained substantial variation in body

Table 2 Results from linear models with logged body mass (in mg), logged wing length (in mm) and logged proboscis length (in mm) as the response variables and family and habitat as predictors

	Estimate mean $\pm$ SE	<i>t</i> -statistic	<i>p</i>
Body mass			
Intercept (Hesperiidae–Ecotone)	4.45 $\pm$ 0.146	30.56	< 0.0001
Lycaenidae	-0.98 $\pm$ 0.169	-5.80	< 0.0001
Nymphalidae	0.43 $\pm$ 0.145	2.97	0.003
Papilionidae	1.34 $\pm$ 0.196	6.84	< 0.0001
Pieridae	0.30 $\pm$ 0.195	1.54	0.124
Habitat Forest	-0.03 $\pm$ 0.119	-0.22	0.826
Habitat Open	-0.62 $\pm$ 0.167	-3.68	< 0.0001
Wing length			
Intercept (Hesperiidae–Ecotone)	2.95 $\pm$ 0.047	62.87	< 0.0001
Lycaenidae	-0.08 $\pm$ 0.054	-1.44	0.152
Nymphalidae	0.56 $\pm$ 0.047	12.07	< 0.0001
Papilionidae	1.02 $\pm$ 0.063	16.13	< 0.0001
Pieridae	0.60 $\pm$ 0.063	9.57	< 0.0001
Habitat Forest	-0.06 $\pm$ 0.038	-1.65	0.101
Habitat Open	-0.25 $\pm$ 0.054	-4.73	< 0.0001
Proboscis length			
Intercept (Hesperiidae–Ecotone)	2.78 $\pm$ 0.068	41.11	< 0.0001
Lycaenidae	-0.97 $\pm$ 0.078	-12.37	< 0.0001
Nymphalidae	-0.41 $\pm$ 0.067	-6.02	< 0.0001
Papilionidae	0.19 $\pm$ 0.091	2.11	0.036
Pieridae	-0.16 $\pm$ 0.091	-1.76	0.08
Habitat Forest	-0.02 $\pm$ 0.055	-0.40	0.691
Habitat Open	-0.09 $\pm$ 0.079	-1.20	0.233

mass ($\log(\text{body mass}) \sim \text{family} + \text{habitat}$; adjusted $R^2 = 0.45$; $F_{6,217} = 31.16$, $p < 0.001$; family: $F_{4,217} = 40.61$, $p < 0.001$; habitat: $F_{2,217} = 8.07$, $p < 0.001$) (Table 2). Open habitat species were smaller than forest and ecotone species (Fig. 1).

Similarly, family and habitat preference explained substantial variation in wing length ($\log(\text{wing length}) \sim \text{family} + \text{habitat}$; adjusted $R^2 = 0.69$, $F_{6,217} = 84.39$, $p < 0.001$; family: $F_{4,217} = 118.75$, $p < 0.001$; habitat: $F_{2,217} = 11.33$, $p < 0.001$) (Table 2). Open habitat species had smaller wing lengths than forest and ecotone species (Fig. 1). Wing length increased allometrically with body mass, but the slope of the relationship varied with family ($\log(\text{wing length}) \sim \log(\text{body mass}) \times \text{family} + \text{habitat}$; $\log(\text{body mass}) \times \text{family}$ interaction: $F_{4,212} = 8.77$, $p < 0.001$) (SI Table 2, SI Fig. 1). Wing length also co-varied with habitat preference, after controlling for body mass ($F_{4,212} = 5.97$, $p = 0.003$) (SI Table 2).

Habitat was not associated with proboscis length, both with and without controlling for body mass (Table 2; Fig. 1; SI Tables 2). Thus, while both body mass and wing length were substantially lower in open habitat species, proboscis length was similar across the three habitat categories. Proboscis length increased allometrically with body mass, but the slope of the relationship varied with family ($\log(\text{proboscis length}) \sim \log(\text{body mass}) \times \text{family} + \text{habitat}$; $\log(\text{body mass}) \times \text{family}$ interaction: $F_{4,210} = 13.31$, $p < 0.0001$) (SI Table 2; SI Fig. 1).

Relationship between functional traits, habitat preference and response to habitat disturbance: Body mass showed a slight positive association with aversion to road (adjusted $R^2 = 0.06$, $Est_{\log(\text{body mass})} = 0.11 \pm 0.054$, $F_{1,47} = 4.14$, $p = 0.048$) (Table 3, Fig. 2). No clear relationship was found between aversion to road and wing length or

proboscis length (Table 3). Forest species showed a greater aversion to road than open or ecotone species (adjusted $R^2 = 0.23$, Est : Ecotone-Open (intercept) = -0.26 ± 0.104 , Forest = 0.47 ± 0.147 , Open = -0.08 ± 0.166 ; $F_{2,42} = 7.49$, $p = 0.002$) (Table 3, Fig. 2). Thus, forest species appeared to prefer the interior while smaller sized and ecotone or open habitat species appeared to be less affected by road.

Aversion to high lantana transects in the forest interior did not detectably vary with wing length, but showed a positive association with both body mass and proboscis length ($\log(\text{body mass})$: adjusted $R^2 = 0.11$, $Est = 0.07 \pm 0.033$, $F_{1,31} = 4.97$, $p = 0.033$; $\log(\text{proboscis length})$: adjusted $R^2 = 0.23$, $Est = 0.22 \pm 0.069$, $F_{1,31} = 10.45$, $p = 0.003$) (Table 3, Fig. 2). Forest species showed a greater aversion to high lantana transects than ecotone-open species (adjusted $R^2 = 0.10$, Est : Ecotone-Open (intercept) = -0.04 ± 0.061 , Forest = 1.17 ± 0.084 , $F_{1,28} = 4.1$, $p = 0.052$) (Table 1, Fig. 2). Thus, high lantana transects were depauperate in species with larger body mass or longer proboscis, and species that prefer forest habitats.

Discussion

Relationship between functional traits, family and habitat preference: Functional traits of organisms may help in understanding many aspects of their ecology, such as ecosystem function, geographic distribution, community assembly and behaviour (Srygley and Dudley 1993; Webb et al. 2010; Hoiss et al. 2012). We observed considerable variation in functional traits of butterflies, which can potentially shed light on their ecology and evolution.

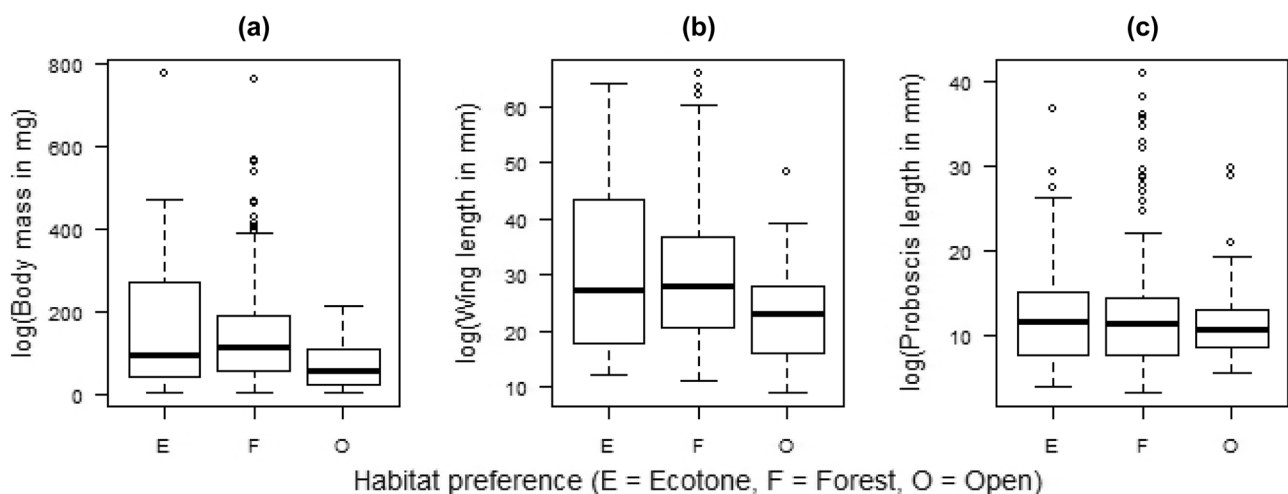


Fig. 1 Boxplots of (a) body mass (b) wing length and (c) proboscis length of butterflies in different categories of habitat preference, i.e., ecotone ($n = 64$), forest ($n = 126$) and open ($n = 37$). In panels (a) and

(b), open habitat species were significantly smaller ($p < 0.05$) than forest or ecotone species; all other categories were not significantly different

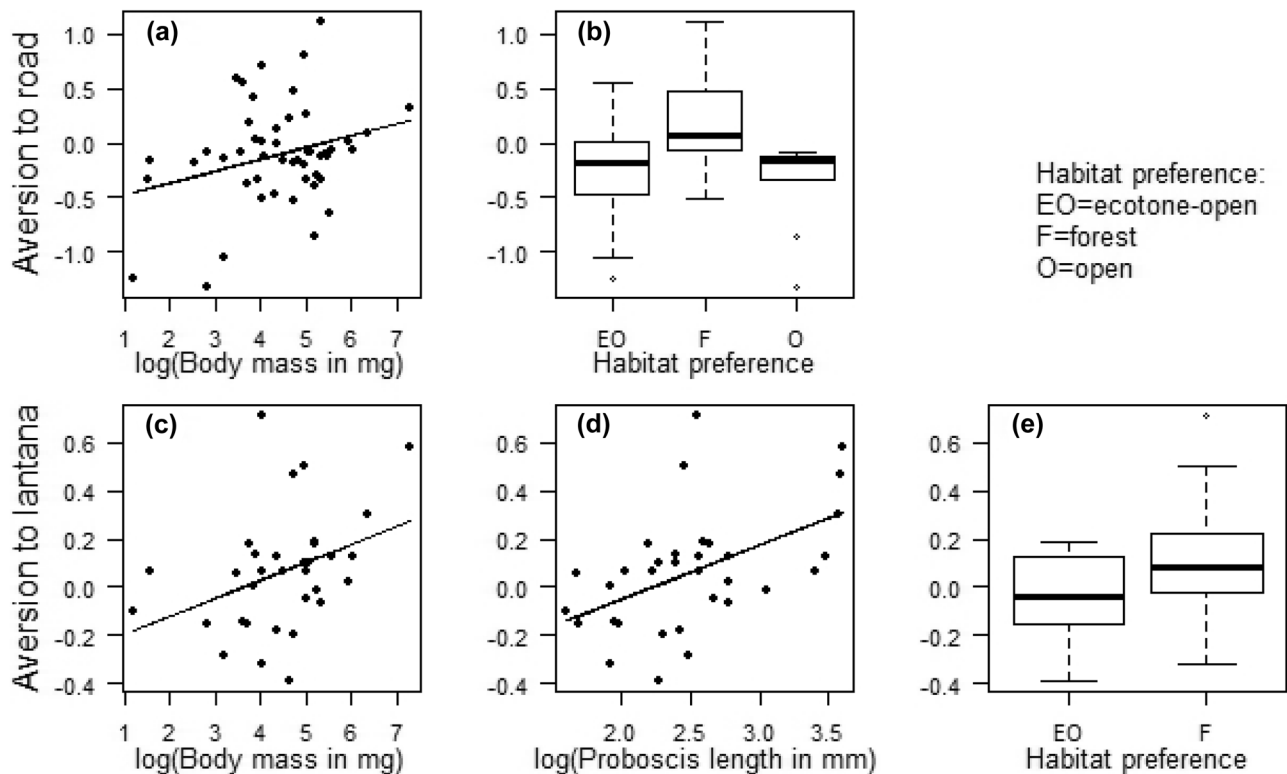
Table 3 Aversion to road and high lantana areas with respect to body mass (in mg), wing length (in mm), proboscis length (in mm) and habitat preference of butterflies. Habitat categories were 'ecotone-

open', 'forest' and 'open' in case of road, and 'ecotone-open' and 'forest' in case of lantana. Positive association implies greater aversion to disturbance

	Aversion to road			Aversion to high lantana		
	Mean (SE)	<i>t</i> -statistic	<i>p</i>	Mean (SE)	<i>t</i> -statistic	<i>p</i>
Body mass						
Intercept	-0.59 (0.247)	-2.40	0.020	-0.27 (0.155)	-1.74	0.092
log (Body mass)	0.11 (0.054)	2.04	0.048	0.07 ± 0.033	2.23	0.033
Wing length						
Intercept	-0.26 (0.472)	-0.54	0.591	-0.18 (0.296)	-0.62	0.540
log (Wing length)	0.05 (0.143)	0.32	0.752	0.08 (0.089)	0.841	0.406
Proboscis length						
Intercept	-0.59 (0.326)	-1.80	0.078	-0.50 (0.178)	-2.80	0.009
log (Proboscis length)	0.20 (0.131)	1.50	0.140	0.22 (0.069)	3.23	0.003
Habitat associations						
Intercept	-0.26 (0.104)	-2.53	0.015	-0.04 (0.061)	-0.70	0.488
Forest	0.47 (0.147)	3.23	0.002	1.17 (0.084)	2.03	0.052
Open	-0.08 (0.166)	-0.48	0.633			

Studies examining association between morphological traits and ecology need to control for phylogeny since morphological traits are an outcome of both historical (phylogenetic) component and recent ecological adaptations (specific

component). Substantial variation in butterfly traits was explained by family associations. In a higher level phylogenetic construction of butterflies, the six butterfly families (i.e., Papilionidae, Hesperidae, Pieridae, Nymphalidae,

**Fig. 2** Aversion to disturbance with respect to butterfly functional traits. Aversion to road with respect to (a) body mass and (b) habitat preference, i.e., ecotone-open (n = 17), forest (n = 17), and open

(n = 11); Aversion to high lantana transects with respect to (c) body mass (d) proboscis length and (e) habitat preference, i.e., ecotone-open (n = 14) and forest (n = 16)

Lycaenidae and Riodinidae) were supported as mutually monophyletic clades (Heikkilä et al. 2012). These phylogenetic relationships, together with the associations we found between family and morphology, imply strong family-level phylogenetic constraints in morphology. Substantial trait variation remained within families, which might be an outcome of interesting adaptive divergences or other evolutionary processes and need to be further examined.

Habitat preference can lead to adaptive divergence and speciation (Berner and Thibert-Plante 2015). Selective forces in a habitat, such as food availability, micro-climate and availability of refuge sites, can lead to adaptive morphological divergence (Goodman 2009; Da Silva et al. 2014). Frog assemblages using similar microhabitats converged in morphology and performance across clades and continents (Moen et al. 2013). In Nymphalid butterflies, species exhibiting polarized wing reflectance were more likely to occupy forest than open habitat, possibly due to the communicative significance of reflectance in the complex light environment of the forest (Douglas et al. 2007). Similar correspondence between phenotype and micro-habitat was found in orb-weaving spiders (Blackledge and Gillespie 2004), “purse case” caterpillars of moth genus *Hyposmocoma* (Kawahara and Rubinoff 2013) and flight morphology of butterflies associated with different micro-environments in the canopy and understorey (Mena et al. 2020). We found clear trait-habitat relations in the pooled dataset of butterflies. Open habitat species were smaller than ecotone and forest species in terms of both body mass and wing length. The results are broadly consistent with previous studies of tropical butterfly assemblages, where larger and darker butterflies (which heat up faster) were associated with cooler habitats such as closed-canopy rainforest or higher elevations, while smaller and lighter butterflies were more common in hotter, open woodlands or lower elevations (Xing et al. 2016, 2018). In our study of habitat preference, ‘forest’ generally refers to hilly or mountainous forested habitats, whereas ‘open’ refers to open canopy dry grasslands, shrublands or other plains areas which tend to be sunnier and hotter. Our findings support Bergmann’s rule, which proposes a general increase in body size in colder climates due to better conservation of heat by larger body sizes.

We couldn’t detect any significant difference in proboscis lengths of butterflies across habitats, suggesting other selective forces not correlated to our habitat categories at play. Butterflies are adapted to feed on a variety of food sources besides floral nectar, such as pollen, bird droppings, rotting fruit, tree sap, wet mud and carrion (Kunte 2000; Krenn 2010). The form and function of mouth parts and feeding behaviour of adult butterflies may show adaptive variation with respect to their feeding guilds or specialized feeding habits (Krenn et al. 2001; Molleman et al. 2005; Krenn 2010; Bauder et al. 2013, 2015a). These evolutionary

mechanisms require further examination in order to explain the variation in proboscis length found in our study, which is the first to quantify such variation in species found in these biogeographical regions.

Relationship between functional traits, habitat preference and response to habitat disturbance: We found butterfly traits to be associated with habitat disturbance. Forest species were more common in undisturbed areas, i.e., the forest interior and low lantana transects, whereas ecotone and open species were more common in disturbed areas. Body mass and proboscis length also showed a positive association with less disturbed areas.

Road provided an open-canopy micro-habitat with a grass and herb understorey (A.S & K. I., unpublished). Lantana was strongly negatively correlated to canopy cover in the forest interior transects (A.S & K. I., unpublished). This could have led to immigration, or an increase/local shift in abundance of sun-loving open habitat butterflies to the road and high lantana areas in the forest interior, whereas shade-loving forest species were more restricted to low lantana transects in the interior. The results support earlier studies of tropical butterflies where habitat degradation (such as road building, selective logging) and warmer temperatures negatively affected shade loving and forest-associated species (Hamer et al. 2003; Bonebrake et al. 2016). Thermoregulation and the relationship between larger body size and heat gain (Xing et al. 2016, 2018) may also partially explain the increased aversion to road and high lantana (both associated with a more open and hence hotter micro-habitat) apparently shown by larger butterflies in our study. The weaker response of body size to road could possibly be due to usage of road for different purposes such as local movement or dispersal by butterflies of different sizes as well as presence of both small (e.g. *Mycalesis junonia*) and large butterflies (e.g. *Melanitis leda*) in the forest interior, perhaps reflecting the heterogeneity of micro-habitats associated with lantana and non-lantana areas. These micro-habitats are likely to be associated with other climatic factors like humidity, larval and adult food resources, vegetation structure, predation etc., all of which are important factors affecting butterfly distribution in tropical forests (Ghazoul 2002; Dumbrell and Hill 2005; Checa et al. 2014; Nyafwono et al. 2015; Cheng et al. 2018; Beirão et al. 2021). The precise mechanisms behind local distributions of butterflies and responses to habitat disturbance require further examination.

Interaction between proboscis length of butterflies (along with other morphological traits such as wing loading) and floral traits such as corolla depth, nectar quality and quantity determines access and preference of butterflies to different floral resources (Corbet 2000; Tiple et al. 2009; Mertens et al. 2021). For species with long proboscis, it is more profitable to feed on deeper flowers with greater nectar reward per flower (Kunte 2007; Bauder et al. 2015b). Higher frequency

of occurrence of species with longer proboscis length in low lantana transects could be due to a preference for patches with deep-flowered forest shrubs or trees (e.g., *Gangara* and *Notocrypta* feeding from flowers of *Costus* and zinzibers), which are more likely to occur in these micro-habitats. Alternatively, the association between long proboscis and low lantana micro-habitat could be an artefact of a positive correlation of proboscis length with body size, which was found to be negatively associated with lantana. Behavioural foraging observations of butterfly species on different floral resources within the habitat and measurements of floral traits and nectar availability would help to disentangle effects of proboscis length and foraging from other factors affecting butterfly responses to non-native plant resources like lantana.

Note that many butterfly species were rare in the datasets and significant unaccounted variation remained in the linear models. Targeted sampling of rare species in future studies should help bring out the patterns of association between traits and habitat disturbance more strongly. This should be complimented with behavioural observations of species and measurements of ecological variables in order to understand the mechanisms behind distribution patterns and their association with traits.

Forest roads and invasive plants can be viewed as causing internal habitat fragmentation, which can potentially change habitat structure, micro-climate, and quality, quantity or configuration of resources. This study provides evidence that even highly mobile taxa such as butterflies may respond predictably to habitat disturbance even at small spatial scales, and that species traits may be useful in deriving general patterns. The results for both road and lantana indicate that these disturbances can potentially change the community structure of butterflies by favouring smaller, ecotone or open habitat species, while populations of larger or forest species might be negatively impacted. Moreover, these traits correspond with broad eco-evolutionary differentiation between species such as body size in association with habitat preference. Traits associated with larger scale ecological patterns such as distribution across habitats can potentially explain responses to habitat disturbances at smaller spatial scales. Our work contributes towards understanding how functional traits can be used to predict shifts in community structure in response to habitat disturbance.

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Declarations

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